MATH 20250
Prof. Lanier

Handout, Day 6

March 31, 2021

Warm-up 1

a) Solve the following system of equations over $\mathbb{R}$:

$3x + 2y = 14$
$7x + 4y = 26$

b) Write the vector $\begin{pmatrix} 14 \\ 26 \end{pmatrix}$ as a linear combination of $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ over $\mathbb{R}$.

During class 1

For each of the following matrices, decide: Is it in row echelon form? is it in reduced row echelon form? When the answer is yes, put a check. When the answer is no, apply row reduction to achieve these forms.

matrix: $\begin{pmatrix} 0 & 1 & 2 & 0 & 3 & 4 \\ 0 & 0 & 0 & 1 & 0 & 3 \end{pmatrix}$ $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 1 & 3 \end{pmatrix}$

row echelon form?

reduced row echelon form?